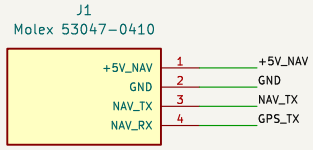
NAV INTERFACE



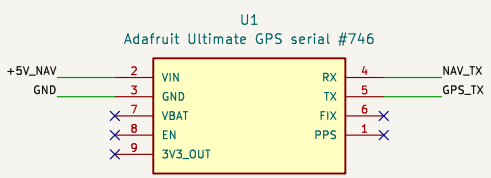
Pin order is a design assumption: verify cable.
TX/RX names are from the main-nav perspective.
UART logic: 3.3 V. Supply: regulated 5 V.

QT PY POWER ISOLATION



GPS VIN connects directly to NAV 5 V.
D1 blocks QT USB power returning to NAV.
QT Py USB-C is for programming.
Power the whole carrier while UART is connected.

U1: COMPLETE GPS BOARD MOUNTED ON TOP

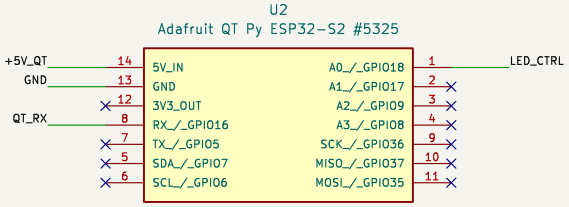


9-pin board-to-board header + two mounting holes.
Serial GPS: no USB connector or flying-wire taps.
NAV TX -> GPS RX; GPS TX -> NAV RX + QT RX.
EN uses onboard pull-up; VBAT uses onboard cell.

DIRECT UART / RECEIVE-ONLY MONITOR

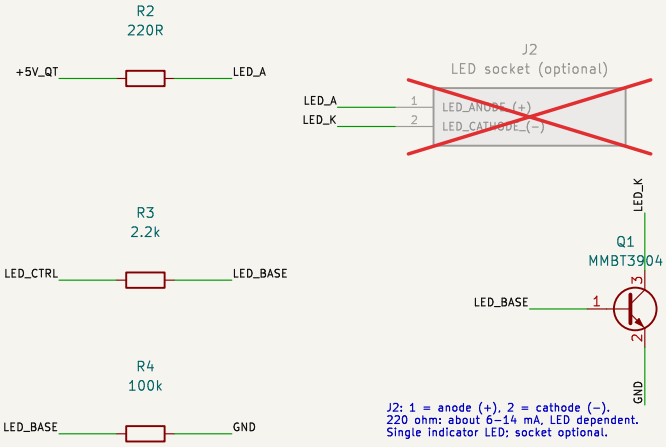


GPS TX feeds J1.4 directly and U2.RX through R1.
QT Py TX is unconnected: monitor only.
No UART driver isolation or GPS rework required.
Header numbering is original Adafruit JP1 order:
1=PPS, 2=VIN, 3=GND, 4=RX, 5=TX,
6=FIX, 7=VBAT, 8=EN, 9=3V3.

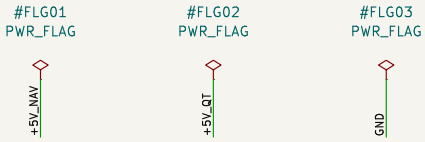


U2: COMPLETE QT PY MOUNTED ON BOTTOM.
Two 7-pin headers; USB-C for programming.
RX = GPIO16; A0 = GPIO18 drives LED.
TX disconnected; disable WI-FI.
Firmware must detect stale/no GPS data.

OPTIONAL STATUS LED - LOW-SIDE DRIVER



J2: 1 = anode (+), 2 = cathode (-).
220 ohm: about 6-14 mA, LED dependent.
Single indicator LED; socket optional.



U1 mounted TOP; U2 mounted BOTTOM

Tern

Sheet: /
File: gps_carrier.kicad_sch

Title: GPS / QT Py two-sided carrier

Size: A4	Date: 2026-10-05	**Rev: B - draft**
KiCad E.D.A. 9.0.7		Id: 1/1